



Unit 4 · Outcome 3 · Scientific Investigation

Generating primary data

KK03

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



You are the data source

00 The big idea You are the data source The exact words of the study design are: “techniques of primary qualitative and quantitative data generation relevant to the selected investigation.” Three ideas are bundled in that sentence, and the exam will test whether you can pull them apart: PRIMARY Whose data is it? Primary data is generated first-hand by you , during your own investigation. QUANT QUAL What form is it in? Numbers and measurements (quantitative) versus descriptions and categories (qualitative). HOW How do you get it? The technique — the instr

Who generated it decides the label

01 Primary vs secondary Who generated it decides the label Primary and secondary are not about the topic or how scientific the data looks — they are purely about who produced it for this investigation . PRIMARY DATA Data you generate yourself as the investigator, by direct measurement or observation during your own method. At Tanderra Creek: the dissolved-oxygen readings you took with a probe; the macroinvertebrates you counted in your net. SECONDARY DATA Data generated by someone else that you access from another source — published studies, EPA Victoria

Numbers vs descriptions

02 Qualitative vs quantitative Numbers vs descriptions Both are completely valid scientific data. The difference is the form the data takes. QUANTITATIVE Numerical data — measurements and counts, always with a unit or a number. Tanderra Creek examples: dissolved oxygen 8.2 mg/L , turbidity 14 NTU , water temperature 13.6 °C , 27 mayfly larvae counted. QUALITATIVE Descriptive data — qualities, categories and observations, not numbers. Examples: water colour “tea-brown”, a faint sewage odour, oily sheen present, substrate = silt, the species identity of an

The instrument bench

03 Techniques of generation The instrument bench A technique is the practical method you use to produce a measurement or observation. Click an instrument on the bench to see what it generates — and whether that data is quantitative, qualitative, or both. click an instrument Field kit. Each tool generates a specific kind of primary data. Select an instrument above Each piece of field kit maps to a technique of primary data generation. Tap one to load it here. Common environmental-science techniques (and the data they generate) Technique What it generates

Technique	What it generates	Data type
Sensor / probe (DO meter, pH probe, EC meter, thermometer, data logger)	A reading on a scale, e.g. DO 8.2 mg/L, pH 7.1, temp 13.6 °C	Quantitative (continuous)
Counting / census (organisms in a net, plants in a quadrat)	A whole-number tally, e.g. 27 mayfly larvae	Quantitative (discrete)
Quadrat (frame placed on ground/streambed)	% cover, counts and a list of species present	Quant + Qual
Transect (line across a gradient)	How a measured variable changes with distance/position	Quantitative
Identification key (macroinvertebrate / plant ID)	The category/species an organism belongs to	Qualitative (nominal)
Field observation / description (colour, odour, sheen, weather)	Described qualities of the site	Qualitative
Photograph / field sketch	A visual record of site condition over time	Qualitative
Turbidity tube	Depth (cm) at which a disk disappears → a number	Quantitative

A single technique can give you both

04 One visit, both data types A single technique can give you both The slickest exam answers recognise that one method often generates both qualitative and quantitative data at once. Picture one kick-sample at Tanderra Creek's downstream site: The quantitative half 27 mayfly larvae, 4 dragonfly nymphs, 61 total individuals, an EPT richness of 3 sensitive taxa. All numbers — all quantitative. The qualitative half Water “tea-brown and turbid”, a faint chemical odour, a thin oily sheen on the surface, fine silt substrate, willow roots overhanging the bank.

The technique must match the dependent variable

05 Choosing techniques that fit The technique must match the dependent variable There's no “best” data type in the abstract — the right technique is the one that actually measures your dependent variable and serves your aim . This is the reasoning examiners reward with the top marks. Match the technique to the data it generates Technique matcher 0 / 5 Check matches Plan a data-collection step Pick the dependent variable you want to capture and the kind of data you need — the planner suggests an appropriate technique. Dependent variable Dissolved oxygen W

Decode what the question is actually asking

◆ COMMAND WORDS

06 Command words Decode what the question is actually asking The verb in the question controls how much you have to write. Tap one to see what it demands for a data-generation question. Identify Describe Distinguish Explain Justify Evaluate

Six ways students lose easy marks

07 Exam traps Six ways students lose easy marks 1 · “Qualitative = unreliable opinion.” Qualitative data is systematic, recorded scientific evidence (species ID via a key, described site condition). It is non-numerical, not low quality. 2 · Calling a downloaded dataset “primary.” If someone else generated the readings, it's secondary — no matter who downloaded or graphed it. 3 · Counts mislabelled as qualitative. A tally is a number → quantitative (discrete). Counting is not describing. 4 · Naming the technique but not the data type. “I used a quadrat” i

Build a top-band answer, live

08 P·E·L writing trainer Build a top-band answer, live Strong short answers follow P·E·L — P oint (name the technique and data type), E vidence (a concrete Tanderra Creek example with units), L ink (tie it back to the dependent variable / aim). Type below and watch the meters respond. “Describe one technique the students could use to generate quantitative primary data for the Tanderra Creek investigation, and identify the data it produces.” (3 marks) Aim for ~50–70 words. Name a technique → give a measured example with a unit → link it to the dependent v

Weak vs strong — same question

09 Worked answer Weak vs strong — same question Question: “Distinguish between qualitative and quantitative data, using one example of each from the Tanderra Creek investigation.” (4 marks) ► Reveal both responses 4 marks X Weak response · 1/4 Quantitative data is good data with numbers and qualitative is just opinions about the creek like if it looks dirty. The students measured the water and wrote down what they saw. No clear distinction stated, no units, “just opinions” misrepresents qualitative data, and the examples are vague — no named variable, no



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Five questions · self-mark as you go

★ PRACTICE & SELF-MARK

10 Practice questions Five questions · self-mark as you go Write your answer, reveal the marking guide, then award yourself marks honestly. Your score builds in the dock below and is saved on this device. Your self-mark · 0 / 20 Work through the five questions above. Be a tough marker — it's how you find the gaps before the real exam. scores

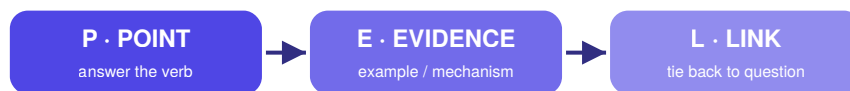
If you remember six things

✦ RECAP / CHEAT SHEET

11 Key takeaways If you remember six things 1 Primary = you generated it. Secondary = someone else did. Downloading doesn't make it primary. 2 Quantitative = numbers (measurements + counts); qualitative = descriptions (qualities + categories). Both are real data. 3 A count is quantitative. Counting larvae produces discrete numerical data — not qualitative. 4 Name the technique and the data type. "I used a quadrat" isn't enough — say what it generates. 5 One technique can yield both qualitative and quantitative data — and the best studies plan for both. 6

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Quantitative data is good data with numbers and qualitative is just opinions about the creek like if it looks dirty. The students measured the water and wrote down what they saw. No clear distinction stated, no units, "just opinions" misrepresents qualitative data, and the examples are vague — no named variable, no number. Barely one mark for the numbers idea.

✓ STRONG / MODEL ANSWER

Quantitative data is numerical , recorded as measurements or counts, whereas qualitative data is descriptive , recorded as qualities or categories rather than numbers. A quantitative example is the dissolved-oxygen reading of 8.2 mg/L taken with a DO probe upstream. A qualitative example is the description of the downstream water as "tea-brown" with an oily sheen . States the distinction (form: numerical vs descriptive), then gives one correctly-typed, investigation-specific example of each — with a unit on the quantitative one. Full marks.